If you've run this code on your own, your network probably looks a little different than what's shown in figure 6.9. That's because network visualizations created with force-directed layouts are the result of the interplay of forces, and, even with a small network like this, that interplay can result in different positions for nodes. This can confuse users, who think that these variations indicate different networks. One way around this is to generate a network using a force-directed layout and then fix it in place to create a network basemap. You can then apply any later graphical changes to that fixed network. The concept of a basemap comes from geography, and in network visualization refers to the use of the same layout with differently sized and/or colored nodes and edges. It allows readers to identify regions of the network that are significantly different according to different measures. You can see this concept of a basemap in use in figure 6.10, which shows how one network can be measured in multiple ways.

Infoviz term: hairball

Network visualizations are impressive, but they can also be so complex that they're unreadable. For this reason, you'll encounter critiques of networks that are too dense to be readable. These network visualizations are often referred to as *hairball*s due to extensive overlap of edges that make them resemble a mass of unruly hair.

If you think a force-directed layout is hard to read, you can pair it with another network visualization, such as an adjacency matrix, and highlight both as the user navigates either visualization. You'll see techniques for pairing visualizations like this in chapter 11.

The force-directed layout provides the added benefit of seeing larger structures. Depending on the size and complexity of your network, they may be enough. But you may need to represent other network measurements when working with network data.

6.2.3 Network measures

Networks have been studied for a long time—at least decades and, if you consider graph theory in mathematics, centuries. As a result, you may encounter a few terms and measures when working with networks. This is only meant to be a brief overview. If you want to learn more about networks, I would suggest reading the excellent introduction to networks and network analysis by S. Weingart, I. Milligan, and S. Graham at http://www.themacroscope.org/?page_id=337.

EDGE WEIGHT

You'll notice that our dataset contains a "weight" value for each link. This represents the strength of the connection between two nodes. In our case, we assume that the more favorites, the stronger a connection that one Twitter user has. We drew thicker lines for a higher weight, but we can also adjust the way the force layout works based on that weight, as you'll see next.